Translations of sine and cosine functions

1

a $y = -1 + \sin x$ **b** $y = -3 + \cos x$

2

a Horizontal translation of 135° left. **b** 135°

3 $y = \sin(x - 30^\circ)$

4 $y = \cos\left(x - \frac{\pi}{6}\right)$

5

a $y = \sin(x + 60^\circ)$ **b** 1

c 360°

6

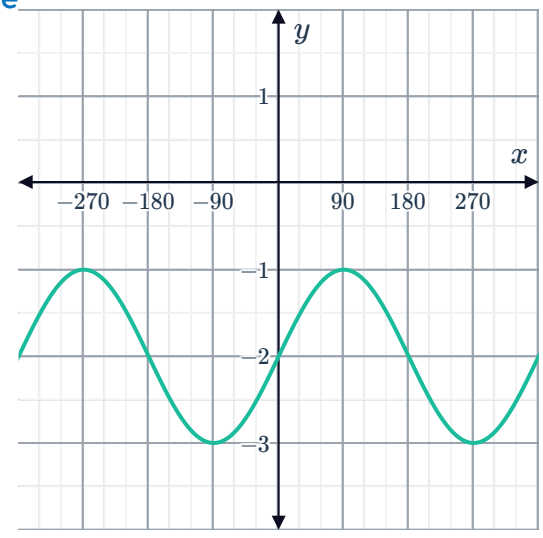
a $y = \cos\left(x + \frac{\pi}{3}\right)$ **b** 1 **c** 2π

7 $-90^\circ, 270^\circ$

8

a 360° **b** 1 **c** -1 **d** -3

e



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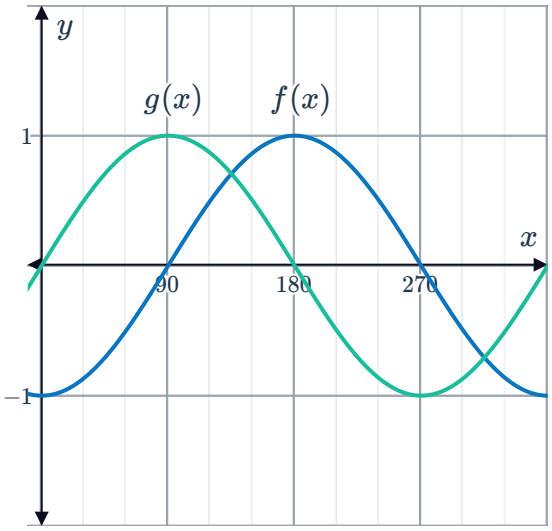
9

a

x	0	90°	180°	270°	360°
$f(x)$	0	1	0	-1	0
$g(x)$	-1	0	1	0	-1

b Horizontal translation 90° to the right

c



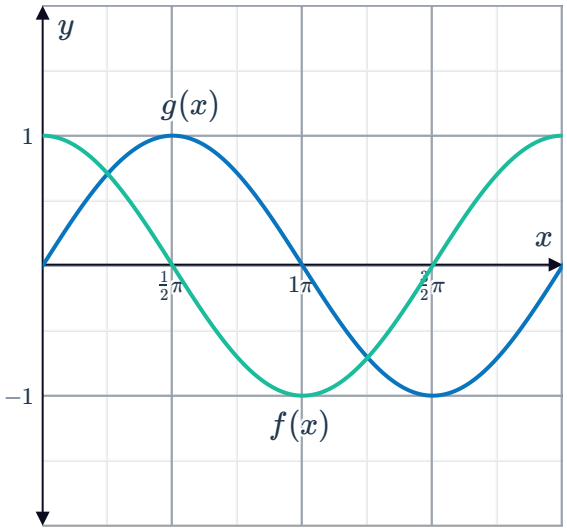
10

a

x	0	$\frac{\pi}{2}$	π	$\frac{3\pi}{2}$	2π
$f(x)$	1	0	-1	0	1
$g(x)$	0	1	0	-1	0

b Horizontal translation $\frac{\pi}{2}$ units to the right

c



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11

a

x	0	$\frac{\pi}{4}$	$\frac{\pi}{2}$	π	$\frac{5\pi}{4}$	$\frac{3\pi}{2}$	$\frac{7\pi}{4}$	2π
$\cos x$	1	0.71	0	-1	-0.71	0	0.71	1
$\cos\left(x+\frac{\pi}{4}\right)$	0.71	0	-0.71	-0.71	0	0.71	1	0.71

b To the left

Dilations of sine and cosine functions

12 $y = 2\cos x$

13 $y = 2\sin x$

14 $y = \cos\left(\frac{1}{2}x\right)$

15

a Yes **b** Yes **c** No **d** No

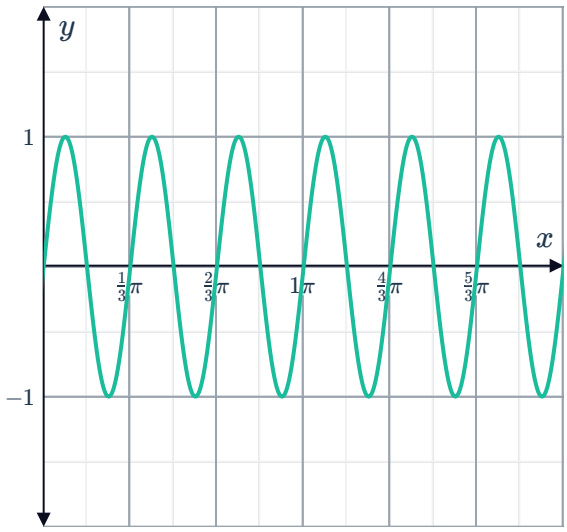
e No **f** Yes **g** No **h** No

i Yes **j** No

16

a $y = 3\sin 6x$

b



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17

a $y = \sin 2x$

b $y = \cos 4x$

18

a 4

b $\frac{1}{9}$

19

a No

b Yes

c No

d No

20

a 5

b -5

c 5

21

a 360°

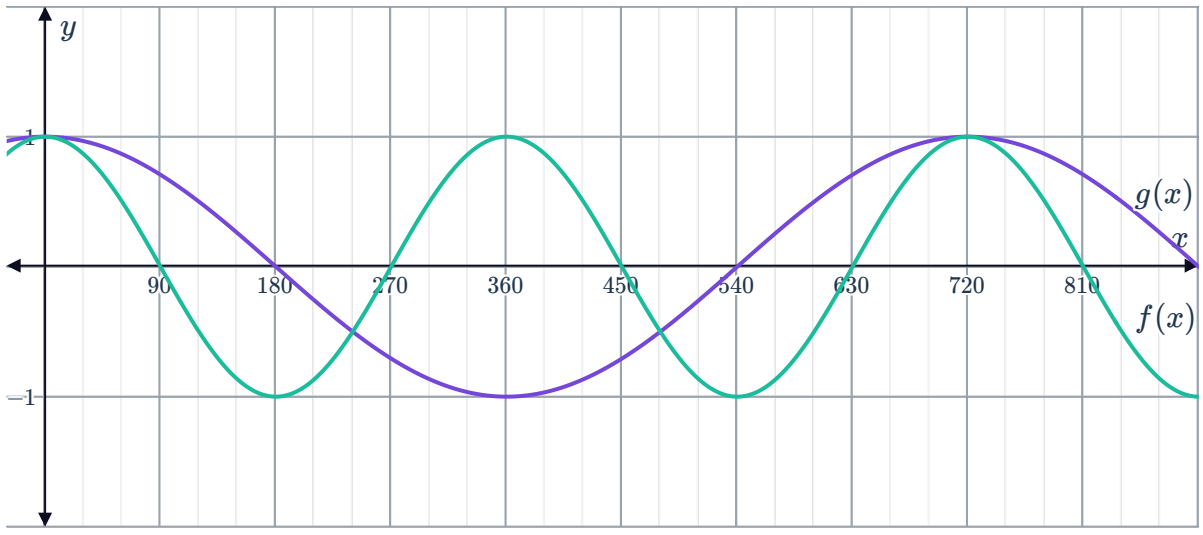
b

x	0°	180°	360°	540°	720°	900°	1080°
$g(x)$	1	0	-1	0	1	0	-1

c 720°

d Horizontal stretch by a factor of 2

e



f No

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Series of transformations

22 The graph of $y = \cos x$ undergoes a phase shift of 45° to the left, and vertically translated downwards by 2 units.

23

a Horizontal stretch by a factor 2 and a horizontal translation (phase shift) of 30° to the left.

b $g(x) = \sin(2x + 60^\circ)$

24

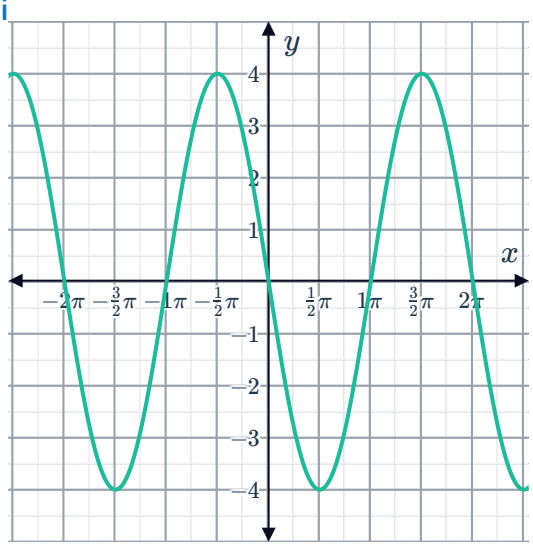
a Horizontal stretch by a factor 3 and a horizontal translation (phase shift) of $\frac{\pi}{9}$ units left.

b $g(x) = \cos\left(3x + \frac{\pi}{3}\right)$ or $g(x) = \cos\left(3\left(x + \frac{\pi}{9}\right)\right)$

25

a

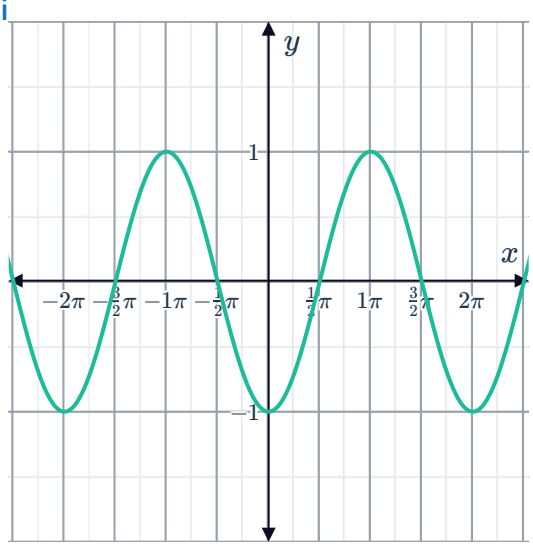
i 4 **ii** π **iii**



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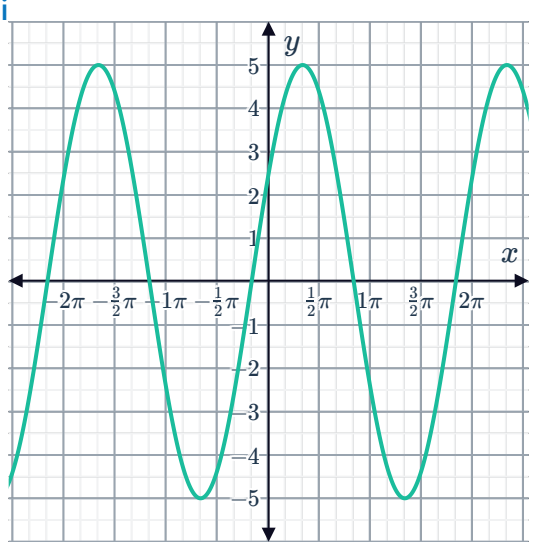
b

i 1 ii $\frac{\pi}{2}$ iii



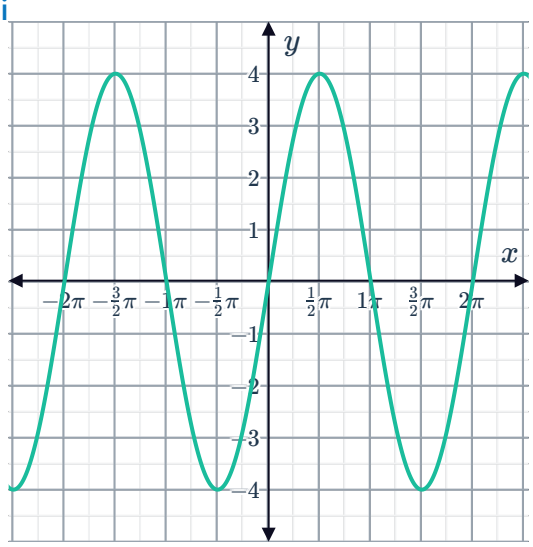
c

i 5 ii $\frac{\pi}{3}$ iii



d

i 4 ii $-\frac{\pi}{2}$ iii

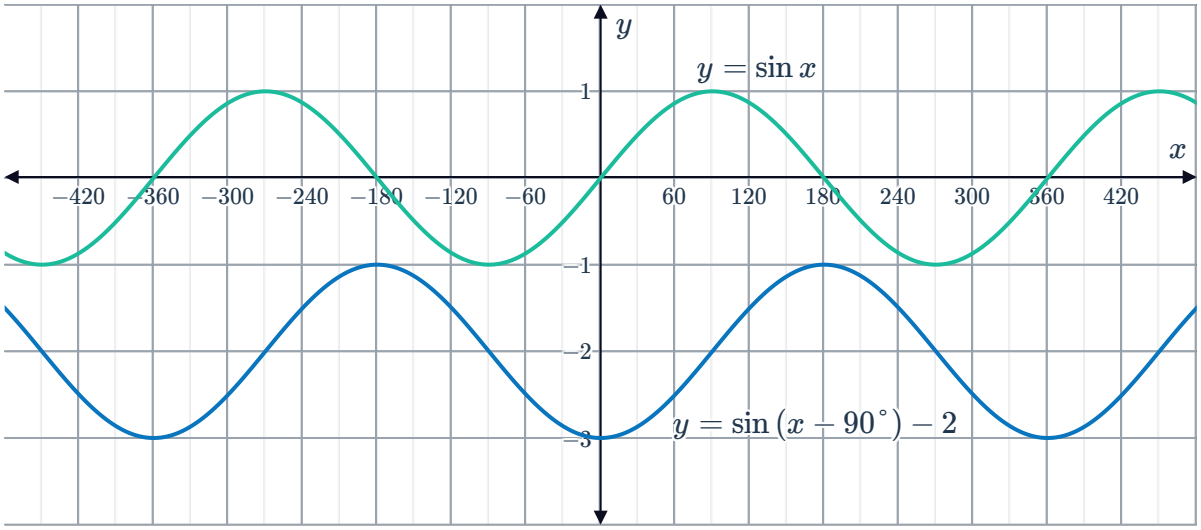


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26

a Move the graph to the right by 90° and down by 2 units.

b

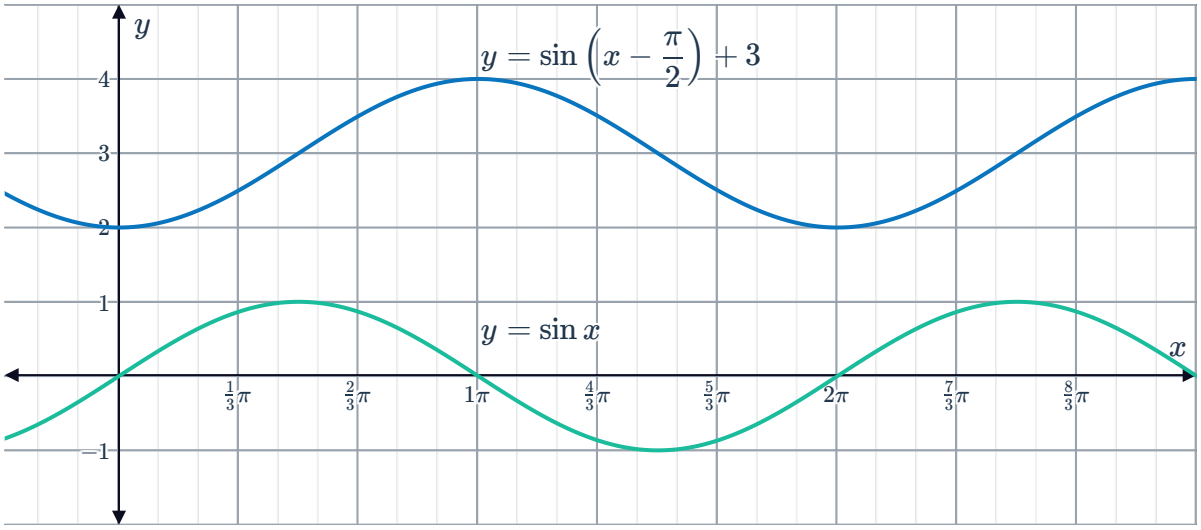


c 360°

27

a Move the graph to the right by $\frac{\pi}{2}$ and up by 3 units.

b



c 2π

28

a $(90^\circ, 5)$ **b** $(270^\circ, 5)$ **c** $(90^\circ, 2)$ **d** $(45^\circ, 1)$

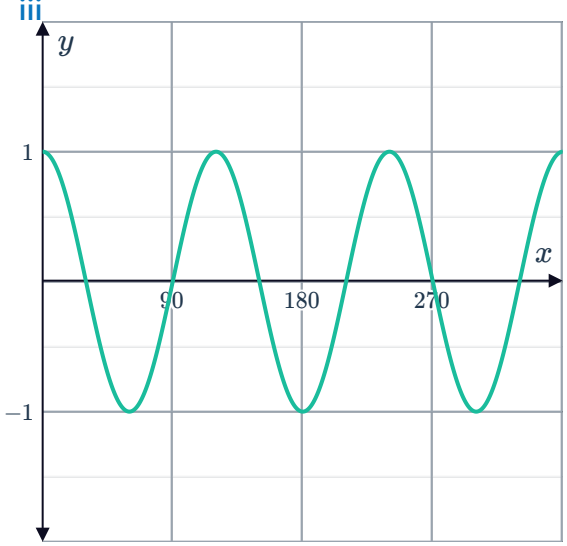
e $(150^\circ, 1)$ **f** $(90^\circ, 6)$

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31

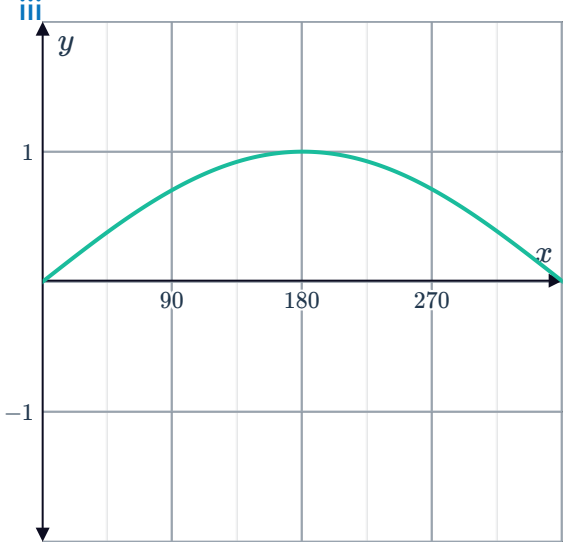
a

i 1 **ii** 120° **iii**



b

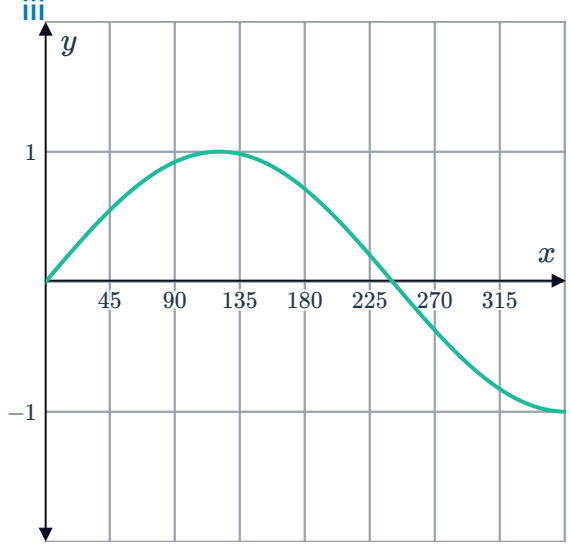
i 1 **ii** 720° **iii**



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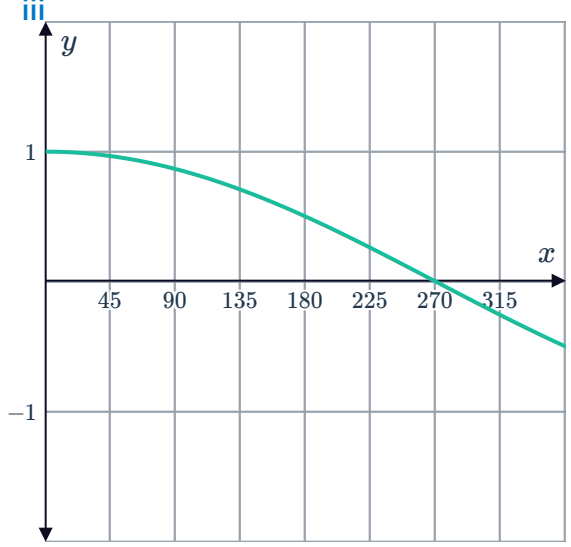
c

i 1 **ii** 480° **iii**



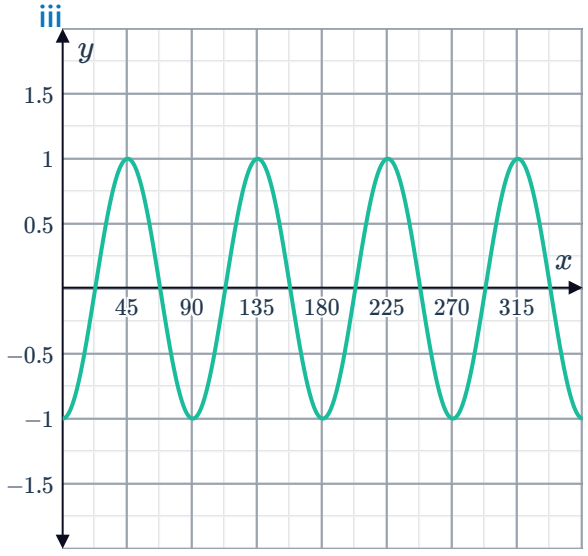
d

i 1 **ii** 1080° **iii**



e

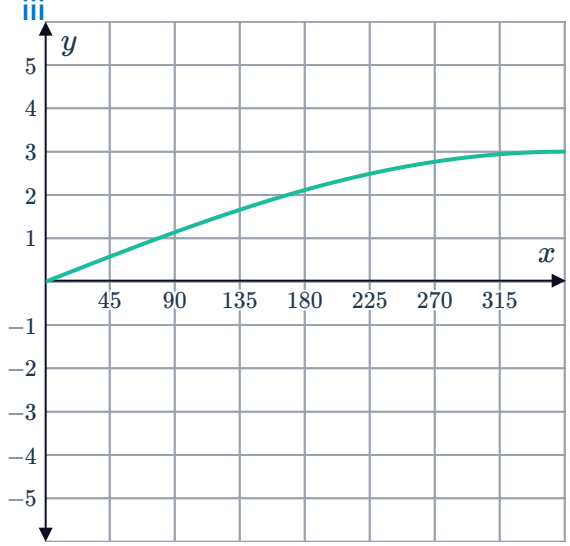
i 1 **ii** 90° **iii**



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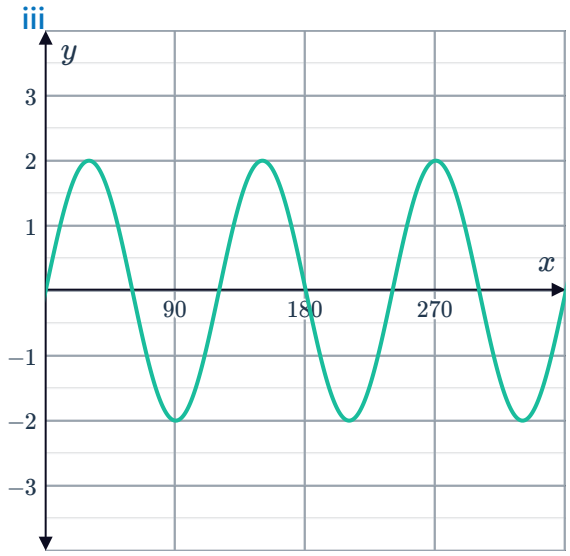
f

i 3 ii 1440° iii



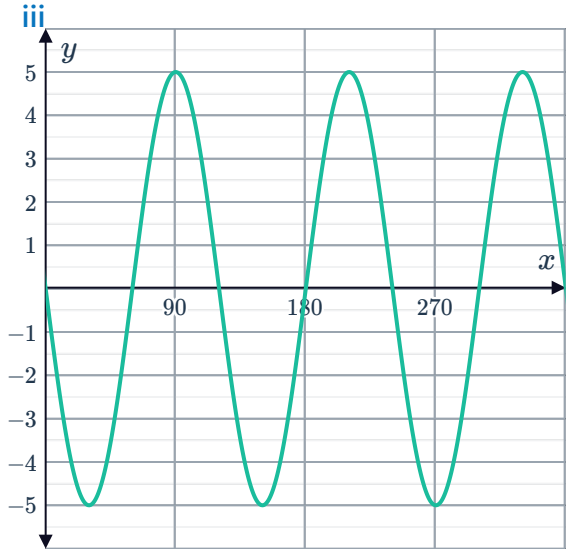
g

i 2 ii 120° iii



h

i 5 ii 120° iii



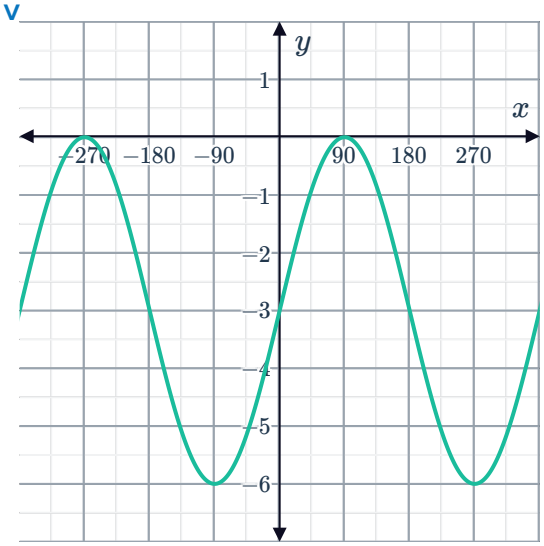
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32

a

i 360° **ii** 3 **iii** 0 **iv** -6

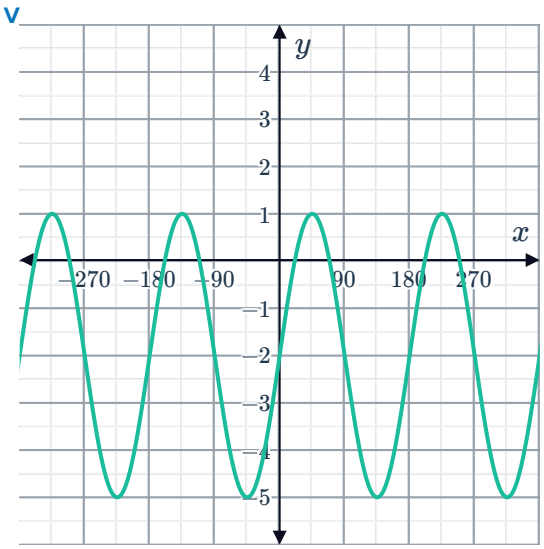
v



b

i 180° **ii** 3 **iii** 1 **iv** -5

v



33

a $y = 3\sin 2x$ **b** $y = -\sin 2x$ **c** $y = 2\cos 3x$ **d** $y = -4\cos\left(\frac{x}{2}\right)$

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Applications

34

a $f(t) = 250 + 100 \sin 2\pi t$

b $g(t) = 350 - 100 \sin 4\pi t$

c 36 times

d $\frac{1}{8}$ years

e $t = \frac{7}{12}$

35

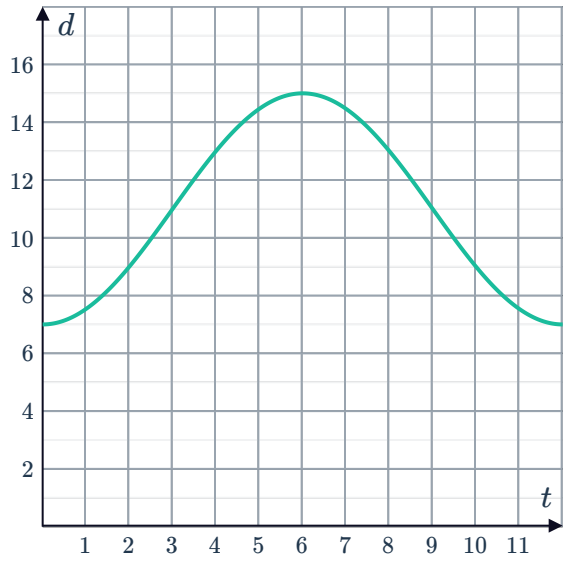
a

Time (t in hours)	0	3	6	9	12
Tide level (d in meters)	7	11	15	11	7

b 4 m

c 12 hours

d



e

i Yes **ii** No **iii** No **iv** No

f $d = -4 \cos \frac{\pi}{6} t + 11$

g 2.66 hours

36

a Sine, the graph's initial value is at the midline.

b $I = \sin\left(\frac{\pi t}{2}\right)$

c 210 times

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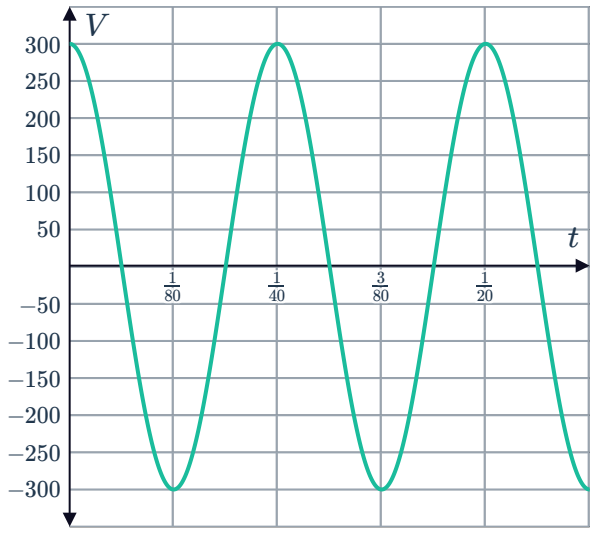
37

a

i Yes **ii** No **iii** No **iv** No

b $V = 300\cos(80\pi t)$

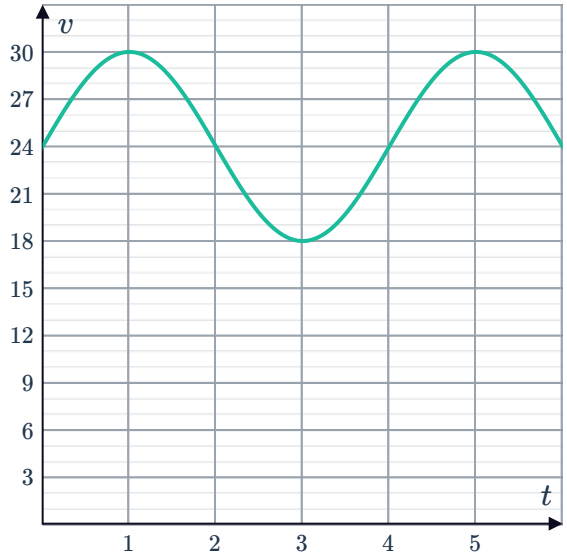
c



d It would decrease the period of the function and compress the graph horizontally.

38

a



b 30 units

c The number of seconds that have passed since the beginning of the section.

d 24 units

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